

Test-Time Failure Prediction for Spacecraft Pose Estimation Under Domain Shift

Stanford CS229 Project

Anjali Sreenivas
Department of Computer Science
Stanford University
anjalisr@stanford.edu

Lundeen Cahilly
Department of Computer Science
Stanford University
lcahilly@stanford.edu

1 Introduction

Autonomous pose estimation of noncooperative spacecraft is essential for safety-critical missions such as on-orbit servicing, debris removal, and satellite refueling (Pauly et al., 2023). Accurate 6D pose estimation from monocular imagery enables safe autonomous rendezvous, proximity operations, and docking, where small errors can lead to collision and mission failure (Park et al., 2022; Pasqualetto Cassinis et al., 2019). These pose estimates feed downstream to navigation filters (e.g., the Unscented Kalman Filter) which rely on per-measurement uncertainty estimates to properly track the target spacecraft’s state. Without reliable per-measurement uncertainty, the filter cannot down-weight poor measurements, degrading navigation performance (Park and D’Amico, 2024).

This challenge is compounded by domain shift. Spaceborne ML models must be trained on synthetic imagery (the only scalable source of labeled pose data), but they are deployed on real images exhibiting harsh illumination, specular reflections, and deep shadows that synthetic renderers struggle to reproduce (Park et al., 2022; Pérez-Villar et al., 2024). The Spacecraft Pose Network (SPNv2) (Park and D’Amico, 2024) is a state-of-the-art estimator that narrows the synthetic-to-real gap, but explicitly identifies uncertainty quantification as unsolved.

Our work addresses this gap by learning to map a compact feature vector, extracted at test time from a single forward pass of SPNv2 on a monocular image, to a reliability estimate for that pose. This estimate is calibrated against a physical hard error threshold (e.g., translation error $< 0.1\text{m}$). We deliberately keep the predictor lightweight, thus adding negligible latency. Without altering SPNv2’s weights or requiring deployment-domain ground truth, our reliability signal allows downstream filters to discard and re-weight unsafe measurements before they corrupt the state estimate.

2 Related Work

Learning-based monocular pose estimation has become the dominant approach for determining the state of noncooperative spacecraft (Pauly et al., 2023; Pasqualetto Cassinis et al., 2019). The KRN (Park et al., 2019) and SPN model families initially established the task on the SPEED/SPEED+ benchmarks (Park et al., 2022), SPNv2 (Park and D’Amico, 2024) is the current state-of-the-art pose estimator with its multi-task heads and Online Domain Refinement, and Pérez-Villar et al. (2024) push domain adaptation further through inter-model consensus. These prior works focus on improving pose estimation accuracy, but none address how reliable a given prediction is at test time.

Standard uncertainty-quantification approaches include temperature scaling (Guo et al., 2017), MC-dropout (Gal and Ghahramani, 2016), and deep ensembles (Lakshminarayanan et al., 2017). However, dropout and ensembles require multiple stochastic forward passes or full network retraining, which is impractical for satellites, which are low SWaP (Size, Weight, and Power) systems where computational resources are highly constrained. We instead extract uncertainty signals from a single forward pass, mirroring the use of raw network outputs for misclassification and out-of-distribution detection (Hendrycks and Gimpel, 2017). To handle the synthetic-to-real domain gap, we explore pairing these

signals with covariate-shift correction, reweighting the training loss by the density ratio between domains (Shimodaira, 2000; Sugiyama et al., 2007).

3 Dataset and Features

Dataset Splits We use the SPEED+ dataset (Park et al., 2022), which comprises 59,960 synthetic images alongside 9,531 hardware-in-the-loop (HIL) images of a known “Tango” target spacecraft. The HIL images are split into two spaceborne illumination domains: diffuse Earth albedo (lightbox) and direct sunlight (sunlamp). Because the SPNv2 model is trained on a subset of the synthetic images, we restrict our analysis to the validation set of 11,994 synthetic, 6,737 lightbox, and 2,787 sunlamp images. The synthetic split provides the only labeled training data for our deployable uncertainty predictors. The HIL ground-truth poses are strictly reserved for evaluation and for fitting the oracle upper bound, since they serve as a proxy for real in-space imagery.

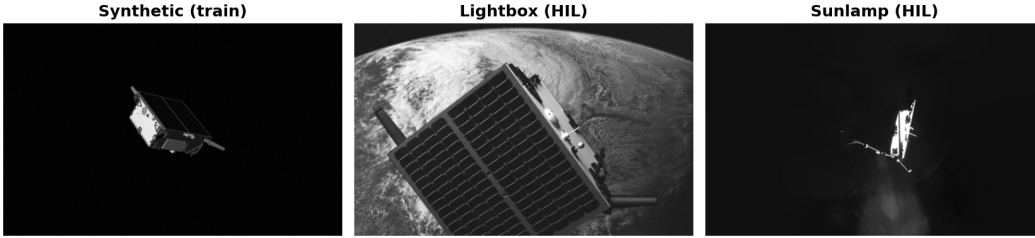


Figure 1: SPEED+ images showing synthetic, lightbox, and sunlamp (HIL) domains.

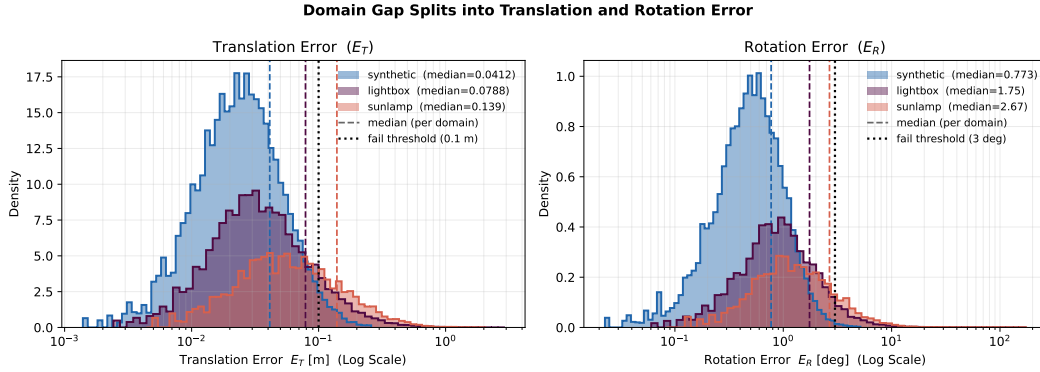


Figure 2: Translation and rotation error across synthetic, lightbox, and sunlamp (HIL) domains.

Domain Gap and Feature Selection The domain gap from synthetic images to the HIL domains is severe (Fig. 1) (Park et al., 2022). Specular glare and deep shadows in the sunlamp condition degrade estimator performance significantly. To ensure our evaluation is relevant to downstream navigation filters, we define prediction failures using absolute physical thresholds rather than domain percentiles. Specifically, we set translation error $E_T > 0.10$ m and rotation error $E_R > 3.0^\circ$. Under these metrics, performance degrades sharply across domains (Fig. 2). From synthetic to lightbox to sunlamp, mean E_T increases from 0.054 m to 0.225 m, driving translation failure rates from 11% (synthetic) to 63% (sunlamp). Rotation suffers as well, with mean E_R climbing from 0.99° to 10.9° .

A robust uncertainty metric must survive the domain gap, which points us toward using features from the model’s own internal state rather than raw image statistics. From a single SPNv2 forward pass, with no ground truth, we extract a 12-dimensional vector of signals in uncertainty/pose space that the network already computes. The features fall into three groups: disagreement between SPNv2’s two pose estimates (direct regression and PnP on predicted keypoints), which diverge when a prediction is unreliable; internal branch uncertainties (heatmap and segmentation statistics, detection score), which reflect localization quality; and geometric covariates (range, bounding-box size), which flag intrinsically hard views. Full descriptions of the selected features are in Appendix Table 3.

Per-feature Spearman ρ vs pose error

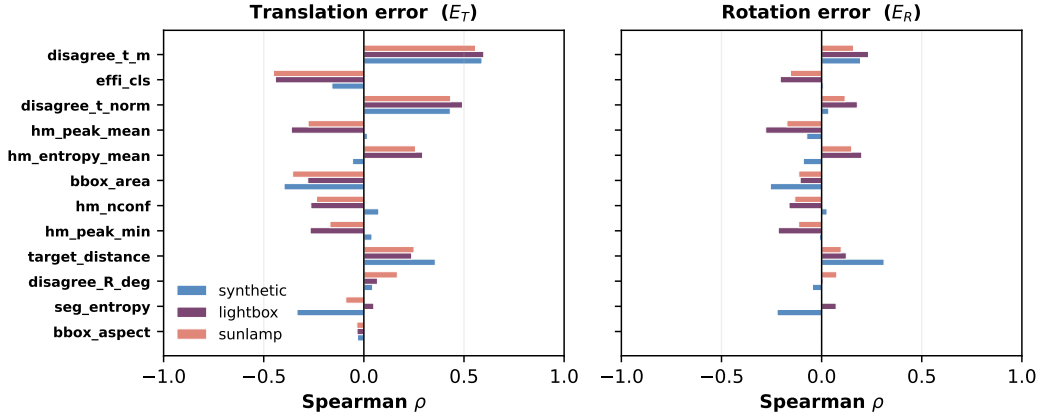


Figure 3: Per-domain Spearman correlation scores between selected features and SPNv2 error.

Per-feature correlation analysis Figure 3 reports the per-domain Spearman correlation between each feature and pose error. Several of the most strongly correlated features for failure signal in the synthetic data (multi-head disagreement (`disagree_t_m`, `disagree_t_norm`), target distance) are consistent across domains, especially for translation error. This domain-invariant failure signal further motivates exploration of this feature space for the uncertainty models that follow.

Preprocessing Raw features span roughly seven orders of magnitude and several are strongly right-skewed, so we apply a $\log(1 + x)$ transform to the four strictly-positive heavy-tailed features and standardize all features to zero mean and unit variance (see Appendix 7.2 for details).

4 Methods

Baselines We evaluate our methods against two baselines. As our minimum performance floor, we first chose a naive random predictor that flags failures with a probability equal to the failure rate of the synthetic training set (11.4% for translation, 2.0% for rotation).

The second baseline thresholds SPNv2’s internal head disagreement as an uncertainty proxy: when the two pose heads diverge, the final estimate is likely unreliable. Since disagreement has no fixed relationship to physical pose error, we cannot threshold it at the failure limit directly. Instead, we select the threshold maximizing F_1 score on the synthetic validation split (translation: `disagree_t_m` ≥ 0.097 , $F_1 = 0.50$; rotation: `disagree_R_deg` $\geq 167.6^\circ$, $F_1 = 0.04$). This forms a strong single-feature signal that our method must outperform to justify the usage of additional features.

Supervised failure predictor (synthetic $\rightarrow$ HIL) We treat translation and rotation as independent failure axes with binary labels

$$y = \mathbf{1}[e > \tau], \quad \tau = 0.10 \text{ m}, \quad \tau = 3.0^\circ,$$

where e is the per-image error. We fit an ℓ_2 -regularized logistic regression on the synthetic data per axis, weighting each example by the inverse frequency of its class to compensate for a low failure rate. The fitted model is applied unchanged to HIL features. Full details are in Appendix 7.3.

Importance-weighted predictor The supervised baseline applies a model trained on synthetic features to HIL features, with no correction for the shift in distribution. To account for this, we reweight each synthetic training example by how much more likely its feature vector x is under the HIL distribution than the synthetic one. We estimate this ratio via a logistic regression domain classifier $d(x) = P(\text{HIL} | x)$, trained on synthetic and HIL features with no class weighting and no failure labels. By Bayes’ rule, the ratio reduces to the classifier odds, $w(x) \propto d(x)/(1 - d(x))$, so synthetic examples with HIL-like features receive higher weight.

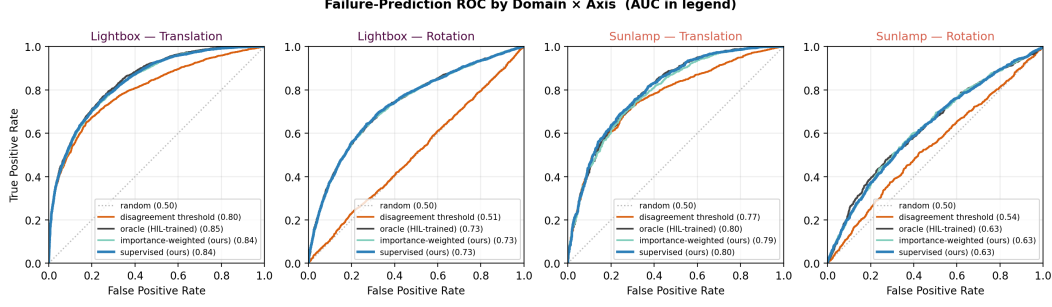


Figure 4: Failure-prediction ROC curves by domain and error axis with per-method AUCs.

We fit $d(x)$ via 5-fold cross-validation so each synthetic image is scored only by a classifier that never trained on it and clip weights at the 99th percentile to prevent a small number of outlier examples from dominating the loss. We refit the class-weighted logistic regression from (2) with per-example weights, giving us the following loss:

$$\mathcal{L}_{\text{IW}}(\theta, b) = - \sum_{i=1}^n w(x_i) \alpha_{y_i} \left[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \right] + \frac{\lambda}{2} \|\theta\|_2^2.$$

This is valid under the covariate shift assumption, which we verify empirically in Appendix 7.4.

Oracle upper bound We fit the same logistic regression approach from the supervised predictor directly on HIL labels, with 5-fold cross-validation to avoid overfitting, to set an upper bound given in-domain supervision. This oracle isolates the cost of the synthetic $\rightarrow$ HIL domain shift.

Unsupervised anomaly detection. Before adopting a supervised approach, we tested whether unsupervised anomaly detection could flag failures without labels, fitting four detectors (Mahalanobis distance, Gaussian mixture, one-class SVM, and isolation forest) on synthetic features and scoring HIL images by how anomalous they appear. The assumption is that failures coincide with off-distribution inputs in the feature space. Some detectors show weak correlation with pose error on HIL domains ($\rho < 0.5$) (Appendix 7.5), but we found supervised prediction substantially more effective.

5 Experiments / Results / Discussion

Experimental Setup and Hyperparameters All classifiers are ℓ_2 -regularized logistic regression (scikit-learn, L-BFGS), and we use the default regularization strength $C = 1.0$. Failure classifiers use `class_weight=balanced`; the domain classifier is left unweighted to preserve calibrated probabilities. We use 5-fold cross-validation in two places: to produce out-of-fold importance weights for the importance-weighted predictor and to evaluate the oracle upper bound.

Evaluation Metrics Failure prediction is binary ($\text{fail} = 1$ if $E_T > 0.10$ m or $E_R > 3.0^\circ$). For each image, the model outputs a score $\hat{p}(x) \in [0, 1]$. Converting this to a binary prediction requires choosing a decision threshold τ . Our primary metric is AUC, the probability that a random failure scores higher than a random success ($\text{AUC} = \Pr(\hat{p}(x^+) > \hat{p}(x^-))$). AUC evaluates ranking quality across all thresholds simultaneously, so it never requires committing to a particular τ . This is important here, since HIL labels are withheld during development, so any metric that depends on a decision threshold would require choosing one arbitrarily or tuning it on synthetic data, both of which could compromise evaluation of the method. For completeness, we also report F_1 /precision/recall of the best deployable model at a fixed $\tau = 0.5$ cut.

Quantitative Results The importance-weighting domain classifiers trained to separate each HIL domain from synthetic achieve high AUC scores of 0.82 (lightbox) and 0.91 (sunlamp), confirming a large, measurable visual gap across domains as we saw in Figure 2.

Table 1 and Figure 4 illustrate the ROC curves and report failure-prediction AUC scores across our methods. Three conclusions follow: (i) SPNv2-internal features are strongly predictive: the

supervised classifier reaches AUC 0.84 on translation error. (ii) Importance weighting does not improve over direct transfer, as the two agree within ± 0.01 AUC, which is well within overlapping bootstrap intervals (Appendix 7.6). (iii) The synthetic-trained model nearly matches the oracle trained on HIL labels (AUC gap ≤ 0.006), indicating it already operates at the in-domain ceiling.

Table 1: Failure-prediction AUC on HIL (0.50 = random). The disagreement baseline uses the SPNv2 head-disagreement score directly (`disagree_t_m` for E_T , `disagree_R_deg` for E_R).

Domain	Error Type	Naive Random	Disagreement	Direct Supervised	IW Supervised	Oracle
lightbox	E_T	0.500	0.800	0.841	0.839	0.846
lightbox	E_R	0.500	0.506	0.734	0.732	0.734
sunlamp	E_T	0.500	0.766	0.802	0.790	0.799
sunlamp	E_R	0.500	0.539	0.627	0.629	0.631

Table 2 gives operating-point metrics for a deployable accept/reject gate: for each axis we choose the F_1 -optimal probability threshold on a held-out synthetic validation split, freeze it, and apply it unchanged to HIL, so no target labels are used to set the operating point. We observe strongest performance on translation (lightbox $F_1 = 0.69$, sunlamp $F_1 = 0.72$) and weaker performance on rotation, mirroring the AUC trend.

Table 2: Operating-point metrics for the direct supervised uncertainty model, evaluated on HIL. The decision threshold is selected per axis to maximize F_1 on a synthetic validation split, then frozen (translation 0.73, rotation 0.91) and applied to HIL.

Domain	Comp.	Threshold	Data Fail Rate	Precision	Recall	F_1
lightbox	E_T	0.73	0.41	0.75	0.63	0.69
lightbox	E_R	0.91	0.27	0.55	0.47	0.51
sunlamp	E_T	0.73	0.63	0.85	0.62	0.72
sunlamp	E_R	0.91	0.45	0.59	0.42	0.49

Qualitative Analysis and Failure Modes. Rotation is predicted far less well than translation across all domains. Dominant domain-invariant features correlate well with translation error, while no single feature carries comparable rotation signal (Figure 3). The synthetic rotation failure rate of only 2% also leaves little for the classifier to learn from. Sunlamp is harder than lightbox (AUC 0.91 vs. 0.82), consistent with its larger domain gap; missed failures are predominantly sunlamp images whose internal features appear unremarkable, suggesting the failures are driven by appearance variation the feature space does not encode. Overfitting risk is low: a small feature space, $n = 11,994$ training examples, and ℓ_2 regularization all constrain model complexity, and the oracle confirms generalization to HIL (Table 1).

6 Conclusion / Future Work

We demonstrate that test-time failure prediction for SPNv2 spacecraft pose estimation is achievable under significant domain shift without modifying base model weights. A logistic regression over 12 internal SPNv2 features achieves AUC 0.84/0.80 (lightbox/sunlamp) for translation and 0.73/0.63 for rotation, outperforming all baselines. Explicit importance weighting provided no additional benefit, and our synthetic-trained models performed on par with an in-domain oracle, confirming that the selected model-internal features capture a robust and domain-invariant failure signal.

Future work should address the rotation performance gap, where sunlamp failures driven by visual ambiguity are not captured by our current feature space. Expanding the feature set is the natural first step. Even in its current form, however, the binary accept/reject signal is already useful: flagging unreliable translation estimates before they enter a navigation filter prevents corrupted state estimates at negligible computational cost. Improving upon this to output a calibrated uncertainty score would enable finer-grained downstream integration, such as dynamically scaling measurement noise in an adaptive UKF to down-weight rather than discard borderline estimates.

7 Appendices

7.1 Feature Set

Table 3 lists the 12 features we extract from a single SPNv2 forward pass, with their Spearman rank correlations against translation and rotation error on the synthetic validation split.

Feature	Description	ρ vs E_T	ρ vs E_R
disagree_t_m	Translation disagreement between estimates (m)	0.58	0.20
disagree_t_norm	Translation disagreement normalized by range	0.46	0.15
disagree_R_deg	Geodesic rotation disagreement between estimates	0.12	0.04
effi_cls	Max EfficientPose detection confidence	-0.45	-0.18
hm_peak_mean	Mean keypoint heatmap peak height	-0.32	-0.23
hm_peak_min	Worst-keypoint heatmap peak height	-0.22	-0.17
hm_nconf	Number of keypoints above detection threshold	-0.25	-0.15
hm_entropy_mean	Mean per-keypoint heatmap entropy	0.28	0.18
seg_entropy	Mean entropy of foreground segmentation mask	-0.02	0.04
target_distance	Predicted range to target $\ t\ $	0.25	0.11
bbox_area	Predicted bounding box area	-0.32	-0.11
bbox_aspect	Predicted bounding box aspect ratio	-0.03	0.00

Table 3: SPNv2 features used as inputs to the confidence predictor, with Spearman rank correlations against translation error E_T and rotation error E_R on the synthetic validation split.

7.2 Preprocessing Details

We initially extracted 13 features but dropped the `reject` flag, which is identically zero on synthetic data and carries no learnable signal, yielding the 12-dimensional vector used throughout. Figure 5 shows the per-feature standard deviation before preprocessing: values span roughly seven orders of magnitude, motivating standardization. We apply a $\log(1 + x)$ transform to the four strictly-positive heavy-tailed features (`disagree_t_m`, `disagree_t_norm`, `seg_entropy`, `bbox_area`), then standardize all features to zero mean and unit variance using statistics computed on the synthetic split.

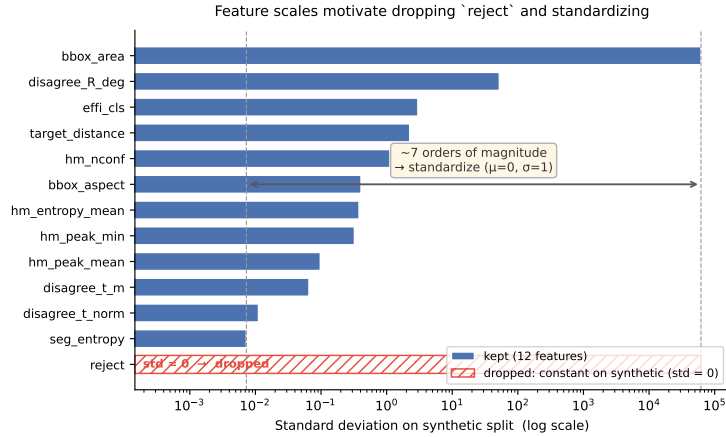


Figure 5: Per-feature standard deviation before preprocessing, motivating standardization. We also drop the `reject` feature, which is constant on synthetic data.

7.3 Supervised Failure Predictor

Per error axis, we minimize the class-weighted cross-entropy with ℓ_2 regularization,

$$\mathcal{L}(\theta, b) = - \sum_{i=1}^n \alpha_{y_i} \left[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \right] + \frac{\lambda}{2} \|\theta\|_2^2,$$

where $\hat{p}_i = \sigma(\theta^\top x_i + b)$ is the predicted failure probability and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid. The per-class weights $\alpha_c \propto 1/n_c$ upweight the rare failure class ($y = 1$) relative to the majority ($y = 0$). The regularization strength is fixed at $\lambda = 1$ (scikit-learn $C = 1.0$), and the fitted parameters (θ, b) are applied unchanged to HIL features at test time.

7.4 Importance-Weighted Predictor

Importance weighting is valid under the covariate shift assumption: $p(x)$ may differ across domains, but the feature-to-failure relationship $p(y | x)$ remains stable (Shimodaira, 2000; Sugiyama et al., 2007), i.e. the same features that predict failure on synthetic data also predict it on HIL. Figure 3 is consistent with this: for the most informative features, correlations with pose error keep the same sign and comparable strength across synthetic, lightbox, and sunlamp. We treat this as a consistency check rather than a proof, since per-feature rank correlations cannot verify the full joint mapping from features to failure.

7.5 Unsupervised Anomaly Detection

We fit four anomaly detectors on standardized synthetic features and scored each HIL image by its anomaly score, testing whether high pose error coincides with off-distribution inputs (Figure 6):

- **Mahalanobis distance:** fits a single Gaussian (μ, Σ) to synthetic features; anomaly score is $(x - \mu)^\top \Sigma^{-1} (x - \mu)$.
- **GMM:** fits a Gaussian mixture via EM; anomaly score is the negative log-likelihood under the mixture.
- **One-class SVM:** fits a decision boundary around the synthetic distribution using an RBF kernel; anomaly score is the signed distance from the boundary. Unlike the parametric methods above, this makes no distributional assumption.
- **Isolation forest:** anomaly score is the average path length required to isolate a point across an ensemble of random trees; points that get isolated quickly are more anomalous.

The signal is present here but relatively weak: translation error correlates up to $\rho \approx 0.47$ (GMM, lightbox) and rotation peaks near $\rho \approx 0.30$. Two issues rule it out as the primary method. First, correlation collapses on synthetic data ($\rho \approx 0$), so the score tracks domain shift rather than per-image failure, and it offers no label-free way to set a threshold on the test domain. Second, even on HIL, it falls short of the supervised predictor (Table 1).

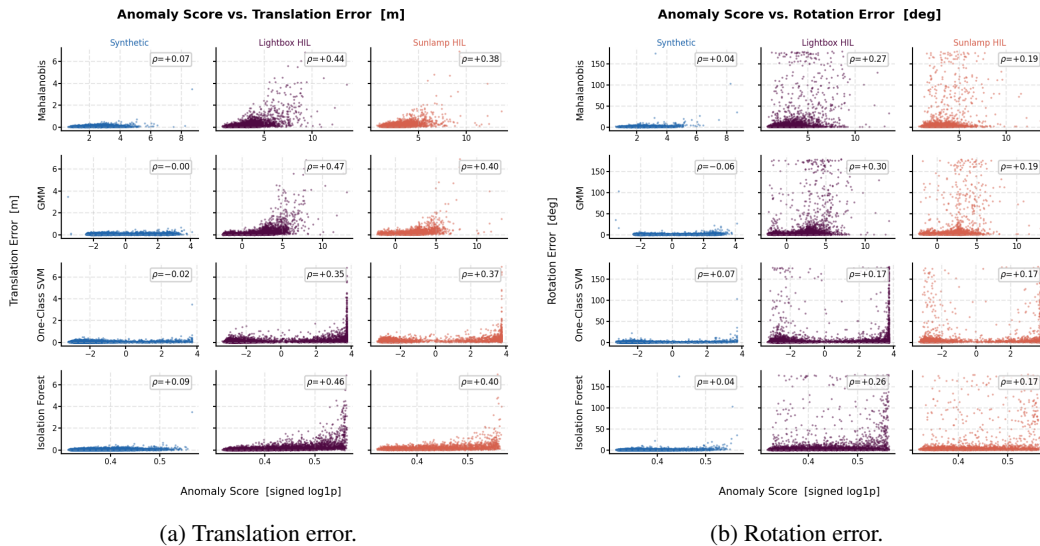


Figure 6: Anomaly score vs. true error across four detectors (rows) and three domains (columns)

7.6 AUC Scores

Figure 7 shows the AUCs from Table 1 as grouped bars, making the baseline-to-ours gap visible at a glance. Each bar’s error bar is a 95% confidence interval obtained by bootstrapping the evaluation set with 2,000 resamples drawn with replacement, reporting the 2.5th and 97.5th percentiles of the resulting AUC distribution.

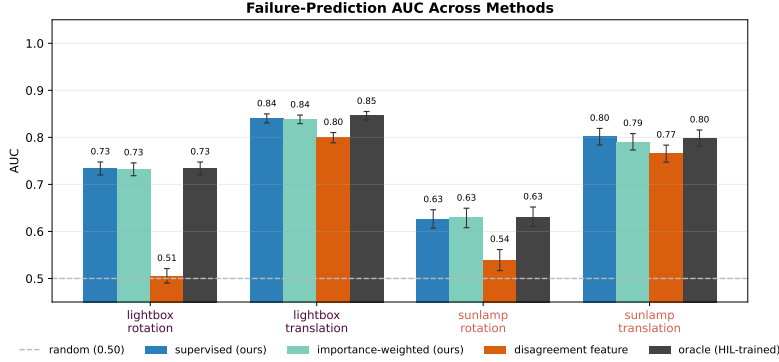


Figure 7: Failure-prediction AUC by method, grouped by domain and error axis.

8 Contributions

Anjali Sreenivas:

- **Pre-milestone:** Ran inference with SPNv2 to collect data and plot pose error distributions; set up and ran milestone baseline experiments; wrote milestone writeup
- **Post-milestone:** Co-developed new methods plan for project, implemented methods, analyzed results, helped validate new feature set, co-wrote paper

Lundeen Cahilly:

- **Pre-milestone:** Researched new project direction (since we decided to pivot from proposal); revised milestone writeup; worked on supervised milestone baseline and analysis
- **Post-milestone:** Researched new feature matrix after running into a roadblock with original plan, ran modified feature-extracting inference on full dataset, co-developed new methods plan for project, implemented baselines and analyzed methods, co-wrote paper

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